

National Neighborhood Data Archive (NaNDA): School Counts by Census Tract, United States, 2000-2018



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nanda_schoolct_tract_2000-2018_01P.dta

nanda_schoolct_tract_2000-2018_01P.csv

nanda_schoolct0018T_01P.sas7bdat

Overview and Data Dictionary

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Dataset Overview

Description

This dataset contains counts of schools per United States census tract from 2000 through 2018. Counts are broken down by type of school (public, charter, magnet, or private) and grade level (elementary, middle, or high).

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Data Sources

Counts of public, charter, and magnet schools from 2000-2018 come from the U. S. Department of Education's Common Core of Data (CCD), the primary database on public elementary and secondary education in the United States. The National Center for Education Statistics (NCES) compiles CCD annually based on data provided by state education agencies (National Center for Education Statistics, n.d.). CCD data was accessed via the Urban Institute's Education Data Portal (Urban Institute, 2020).

Private school counts come from the Private School Universe Survey (PSS). PSS is administered every two years by the National Center for Education Statistics and is completed by private school administrators. Data for 2005-2018 was obtained from the NCES website (National Center for Education Statistics, n.d.).

The 2010 TIGER/Line shapefiles for census tracts (U. S. Census Bureau, 2010) were used to assign schools to tracts.

Coverage

The dataset contains one observation per year per census tract in the United States, including U. S. island territories. Data on private schools is missing for the U. S. island territories.

Methodology

This dataset highlights the concentration of K-12 public, charter, magnet, and private schools per census tract in the United States. Along with measures of school district characteristics and neighborhood school gap (Kim et al., 2021, Gomez-Lopez et al., 2021), it aims to contribute to the discussion about the causes and the consequences of variations in educational context across geographies. We view educational context as a production of the constant interplay of diverse demographic, institutional, and political factors influencing resource distribution and education quality (Arum, 2000). Thus geographic variations in education may play a distinct role in (re)producing social and educational inequalities (Owens, 2020).

In this dataset, we present the count of schools by types at each census tract drawn from school-level data on all public schools and private schools reported by administrators for selective years in the United States. School count measures can be studied as either outcome

or exposure. Researchers may straightforwardly evaluate the potential education policy impact on or the association between neighborhood ethnoracial and socioeconomic context and the change of school counts (Barzil & Candipan, 2021). School site, along with other neighboring places, is also proposed as a site for social inclusion (Wright et al., 2020) and they may potentially benefit neighborhood residents' health and well-being.

To create this dataset, we used the Common Core of Data (CCD) within the Urban Institute's Education Data Portal API to obtain the following school-level data on all public schools in the United States:

- Latitude and longitude of school
- Lowest and highest grade offered
- Whether or not the school is a charter school
- Whether or not the school is a magnet school

We assigned public schools to 2010 census tracts using the latitude and longitude found in CCD. We used the charter school and magnet school flags to create counts of traditional public, public charter, and public magnet schools within each census tract. Note that charter schools are a subset of all public schools. For example, in a tract containing two traditional public schools and one charter school, the variable `public_school` would be three, not two. Note also that a school may be considered both a charter school and a magnet school. As a result, the total number of charter and magnet schools may exceed the total number of public schools in a tract.

For private schools, we obtained addresses and lowest/highest grades from the NCES Private School Survey (PSS). Since the survey is collected every two years, we used data from each two-year academic year range to represent both years (for example, data for 2005 and 2006 comes from the survey collected for the 2005-2006 academic year). We geocoded private schools' addresses from the PSS and assigned those to 2010 census tracts.

For both public and private schools, we used each school's highest and lowest grade offered to classify the school as an elementary, middle, and/or high school. The algorithm for determining how to classify a school was as follows:

- Elementary school = lowest grade < 5
- Middle school = lowest grade < 8 + highest grade > 6
- High school = highest grade > 9

We then used these classifications to create counts of all public, traditional public, public charter, public magnet, and all private elementary, middle, and high schools. Note that one school may count toward more than one classification. For example, a public school offering grades K-8 would count as both a public elementary school and a public middle school. A tract containing one charter school offering grades K-8 would have a value of 1 in the variables `public_charter`, `public_charter_elem`, and `public_charter_middle`.

Missing/zero values are handled differently in this dataset for public and private schools due to differences in data sources. If no public schools are found in a tract, we represent the count as zero because the Common Core of Data is comprehensive and response is mandatory. The Private School Survey is recommended but not mandatory. If no private schools are found in a tract, it may mean there are none, or that school administrators did not respond. Because of this uncertainty, we represent zero values as missing data.

Related Datasets

Data users interested in a broader picture of a neighborhood's school and educational landscape might find useful data in these other NaNDA datasets:

- [School demographics and funding by school district, United States, 2000-2018](#) (Kim et al., 2021).
- [Neighborhood school gap by census tract, United States, 2009-2010 and 2015-2016](#) (Gomez-Lopez et al., 2021).

Variables

Variable	Type	Obs	Unique	Mean	Min	Max	Label
tract_fips10	string	903655	58111	.	.	.	Census tract FIPS code, 2010 TIGER/Line shapefiles
year	int	903655	19	2009.317	2000	2018	Year
all_school	byte	903655	48	2.306558	1	59	Total schools (public & private, all grade levels) in tract
elem	byte	903655	34	1.457295	0	40	Elementary schools (lowest grade < 5) in tract
middle	byte	903655	30	0.814277	0	37	Middle schools (highest grade > 6, lowest grade < 8) in tract
high	byte	903655	32	0.755026	0	46	High schools (highest grade > 9) in tract
public_school	byte	903655	48	2.013817	1	59	Total public schools (traditional public, charter & magnet) in tract
public_trad	byte	903655	51	1.851261	0	59	Traditional public schools (non-charter, non-magnet) in tract
public_charter	byte	903655	17	0.107835	0	25	Charter schools in tract
public_magnet	byte	903655	8	0.056409	0	7	Magnet schools in tract
pub_elem	byte	903655	34	1.228438	0	40	Public elementary schools in tract
pub_middle	byte	903655	22	0.613617	0	25	Public middle schools in tract
pub_high	byte	903655	31	0.526615	0	46	Public high schools in tract
public_trad_elem	byte	903655	36	1.131375	0	40	Traditional public elementary schools in tract
public_trad_middle	byte	903655	22	0.539004	0	25	Traditional public middle schools in tract
public_trad_high	byte	903655	31	0.469509	0	46	Traditional public high schools in tract
public_charter_elem	byte	903655	16	0.065804	0	19	Charter elementary schools in tract
public_charter_middle	byte	903655	13	0.056451	0	16	Charter middle schools in tract
public_charter_high	byte	903655	15	0.042471	0	17	Charter high schools in tract
public_magnet_elem	byte	903655	6	0.032261	0	5	Magnet elementary schools in tract
public_magnet_middle	byte	903655	5	0.019075	0	4	Magnet middle schools in tract
public_magnet_high	byte	903655	7	0.015318	0	6	Magnet high schools in tract
private_school	byte	193979	29	1.36374	1	34	Total private schools in tract
private_elem	byte	193979	11	1.066136	0	16	Private elementary schools in tract
private_middle	byte	193979	30	0.934776	0	34	Private middle schools in tract
private_high	byte	193979	30	1.064058	0	34	Private high schools in tract

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